Cross-Scene Classification of Hyperspectral Images via Generative Adversarial Network in Latent Space

Yahan Yang, Yang Xu, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Biqi Wang, and Zhihui Wei, *Member, IEEE*

***Abstract*— Classifying high-dimensional hyperspectral image (HSI) with limited labeled samples is a difficult problem. One effective solution is to leverage knowledge from scenes with well-labeled images (the source domain) to aid training in the target domain. However, since the source and target domains have different category spaces, it is crucial to extract more discriminative features and address domain adaptation challenges. To tackle this issue, we propose a cross-scene classification method for HSIs via generative adversarial networks (GANs) in latent space (GLSs). Our method employs autoencoders (AEs) to map the input data to a latent space, where the most effective feature representation is extracted and preserved by deep residual 3-D convolutional neural networks (CNNs). The unlabeled samples in the target domain are also utilized in the AE, which ensures that all the samples are considered. We leverage conditional adversarial domain adaptation to overcome the domain shift and introduce maximum mean discrepancy loss to minimize distribution differences between the two domains, facilitating better domain distribution alignment. We tested our approach on three public datasets and demonstrated that it outperforms existing few-shot learning methods. Our results highlight the effectiveness of our classification method via GLSs for HSIs and show that it has potential for practical applications.**

***Index Terms*— Autoencoder (AE), domain adaptation, few-shot learning (FSL), generative adversarial network (GAN), hyperspectral image (HSI) classification, latent space, maximum mean discrepancy (MMD).**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D images, which contain many continuous bands and have a lot of spectral and spatial information [1]. Due to its richness and multidimensionality of information, it plays a crucial role in numerous fields [2], [3], [4], [5], [6], including, but not limited to, environmental monitoring and disaster prevention. As an important application, HSI classification has attracted extensive attention in recent years.

Traditional HSI classification techniques, such as K-nearest neighbor (KNN) [7], logistic regression [8], random forest [9], and support vector machines (SVMs) [10], have been widely used and can yield satisfactory results. However, these methods often overlook the spatial adaptability of spectral features, which is a crucial aspect of HSI analysis.

Deep learning has demonstrated remarkable proficiency in extracting information-rich features. Its capability to learn features automatically has considerably enhanced the flexibility of model application scenarios [11], [12], [13]. As a result, deep learning has witnessed widespread adoption in various image-processing tasks in recent years. Chen et al. [14] proposed a deep feature extraction method based on stacked autoencoder (SAE) [15], with principal component analysis (PCA) [16], deep learning architecture, and logistic regression as the main structure. This approach entails constructing a novel deep learning framework, where logistic regression is applied for the final classification stage.

Nevertheless, the training of deep learning models typically demands an ample amount of labeled data. In the realm of remote sensing, gathering a sufficient quantity of labeled data is often a time-consuming and laborious process. Consequently, the limited availability of training samples directly contributes to the deterioration of classification performance. To address this challenge, various approaches have been employed, including semisupervised [17], [18], [19] and unsupervised feature learning methods [20], [21]. These methods aim to alleviate the dependence on labeled data by leveraging additional unlabeled or partially labeled data to enhance the feature learning process. Liu et al. [22] proposed an HSI classification method based on active deep learning [23], [24], [25]. They employed fully connected networks to perform classification tasks on HSIs. The training process of the network was divided into unsupervised pretraining and fine-tuning stages, with the latter utilizing labeled data. During the pretraining phase, the proposed method involves mapping samples from the unlabeled dataset into an intermediate feature representation. Subsequently, the middle feature is reconstructed through a decoding procedure, and the weights of each layer are learned by improving the similarity of the reconstructed samples to the original ones. However, it is worth noting that this method may become less efficient when the network depth increases.

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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing 210094, China (e-mail: yangyahan@njust.edu.cn; xuyangth90@njust.edu.cn; zebin.wu@gmail.com; wangbiqi@njust.edu.cn; gswei@njust.edu.cn).

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While semisupervised and unsupervised methods help mitigate the challenge of limited labeled data, they rely on the assumption that similar data are sampled from the same distribution. To address the issue of few labeled samples more effectively, the cross-scene classification method has been proposed. This approach utilizes the specific information of related classes from a scene with an adequate number of labeled samples (called the source domain) to aid in training the HSI scene with limited labeled samples (called the target domain). Two primary approaches are commonly employed to tackle the cross-domain learning task: domain adaptation-based methods [26], [27], [28], [29], [30] and fine-tuning-based methods [31], [32], [33].

To enhance the accuracy of cross-scene classification and improve generalization to invisible classes during the training process, the concept of "few-shot learning (FSL)" [34] has emerged. The FSL method can be seen as a form of metalearning, where the acquired general information (known as metaknowledge) from the source class data aids in making predictions about the target class data, even when there is limited availability of labeled target class data. By leveraging this metaknowledge, FSL enables the model to generalize and infer patterns from a few labeled examples, thus facilitating accurate classification of unseen classes. Liu et al. [35] proposed a deep FSL (DFSL) method for HSI classification. This approach employs a deep residual network [36] based on the 3-D convolutional neural network (CNN) to learn the metric space using four source class datasets. During the testing phase, SVM and nearest neighbor methods are utilized to classify the unlabeled data within the test dataset.

While FSL methods have demonstrated the ability to enhance classification accuracy with limited labeled test data, they often overlook the challenge of domain shift between domains. In addition, these methods may not effectively utilize the information available from the labeled target class data. Generative adversarial networks (GANs) have proven to be effective in addressing the domain adaptation problem by extracting domain-invariant features through adversarial training between two networks. In recent years, GANs have gained significant popularity in the field of HSI classification. Several notable GAN-based methods have been developed, including adaptive dropblock-enhanced GAN (ADGAN) [37], progressive growing GAN (PG-GAN) [38], GAN based on collaborative learning and attention mechanism (CA-GAN) [39], adaptive weighting feature fusion GAN (AWF^2-GAN) [40], and spectral–spatial attention feature extraction GAN (SSAT-GAN) [41]. These approaches leverage GANs to enhance HSI classification by leveraging their generative and discriminative capabilities, and incorporating attention mechanisms and feature fusion techniques. Li et al. [42] proposed a novel deep cross-domain FSL (DCFSL) method that addresses the challenges of FSL and domain adaptation within a unified framework for the first time. This method combines transfer learning and deep learning techniques, and expands the available training set by incorporating source-domain samples. To overcome domain shift, a conditional adversarial domain adaptation strategy is employed, aligning the domain distributions and effectively resolving the few-shot problem under domain shift. DCFSL represents a pioneering effort in integrating FSL and domain adaptation within a single framework.

Building upon the aforementioned discussion, we propose a novel cross-scene classification method of HSIs via GAN in latent space (GLS). This method combines the advancements in CNN, deep learning models, and transfer learning to achieve improved cross-scene classification performance. By harnessing the capabilities of GANs and the insights from these research areas, our approach aims to enhance the accuracy and robustness of HSI classification in cross-scene scenarios. First, our proposed approach involves mapping the data from both the source and target domains into a latent space using an autoencoder (AE). This mapping process aims to reduce the feature dimensions of the input data, enabling us to obtain sample inputs with consistent dimensions in the latent space. Next, features are extracted from the target-domain samples and source-domain samples using an embedded feature extractor. We employ a distance measurement method based on maximizing domain invariance to calculate the distance between the source and target distributions. This step helps in narrowing the differences between the two domains, ensuring better alignment and compatibility between them. The spatial–spectral metric space is learned by extracting classification-related features, and then, FSL is performed by comparing the distance between labeled and unlabeled data. The AE and the deep residual 3-D CNN are trained in a unified framework, and the FSL is executed in both the source and target domains. The contributions of this article can be summarized as follows.

1) A novel network based on GLS is proposed to realize cross-scene classification of HSIs. Our proposed network utilizes AEs to map the input data into a latent space and employs a deep residual 3-D CNN in the latent space for feature extraction. This approach allows us to extract and retain the most effective feature representations, facilitating feature alignment while reducing the loss of information. By extracting features from this module, we ensure the consistency of features after projection while preserving their discriminative characteristics. This allows for the generation of robust and informative feature representations, thereby enhancing the overall performance of the classification task.
2) We proposed an adversarial module that leverages conditional adversarial domain adaptation to overcome domain shift. In addition, we introduced the maximum mean discrepancy (MMD) loss to minimize distribution differences between the two domains, facilitating better domain distribution alignment. By incorporating these techniques, we aim to boost a more effective domain adaptation process and enhance the alignment of the source and target domains.
3) A series of experiments and comparisons demonstrate that our proposed method outperforms existing FSL methods by significantly improving classification accuracy with a small number of labeled samples. Furthermore, an ablation experiment has verified the effectiveness of the introduced MMD loss.

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The remainder of this article is organized as follows. In Section II, some previous knowledge and works related to this issue in HSI are briefly reviewed. Section III describes the proposed method in detail. The experiment results and analysis of the proposed network and the comparison method on three datasets are presented in Section IV. Finally, conclusions of our work are drawn in Section V.

II. Related Work

A. Transfer Learning

1) Definition: While traditional machine learning techniques have achieved considerable success in practical applications, they often rely on having a substantial number of labeled instances that closely match the distribution of the test data. However, obtaining sufficient labeled instances can be challenging in many scenarios. Consequently, cross-domain learning has emerged as an effective approach to address this issue.

Transfer learning is the focus of cross-domain knowledge transfer, which aims to improve learning performance by using the knowledge of the related domain (source domain) or to minimize the number of labeled samples required by the target domain. On the basis of given source domain D_S and source task T_S, and target domain D_t and target task T_t, the goal of transfer learning is to use the knowledge of D_S and T_S to improve the prediction effect of learning function of target task (under the condition that D_S is not equal to D_t or T_S is not equal to T_t) [43].

As depicted in Fig. 1, transfer learning, as illustrated in Fig. 1(b), differs from traditional machine learning tasks shown in Fig. 1(a). In transfer learning, the learning process of a new domain is facilitated by leveraging knowledge gained from previous tasks, including data features and model parameters. This enables the new domain to establish its own model by building upon the knowledge and insights acquired from the source tasks. Transfer learning serves as a bridge that transfers and adapts relevant knowledge, facilitating the efficient and effective learning process in the target domain.

Transfer learning can enhance the learning efficiency of the target domain by leveraging data from the source domain. On the one hand, it addresses the issue of limited labeled data, and on the other hand, the knowledge of the pretrained model enhances the generalization ability of the target task. In addition, transfer learning expedites the learning process in the target domain by capitalizing on the knowledge and experience obtained from the source domain. By utilizing learned features as a foundation for the initial model, the training time required for the target task can be effectively reduced. However, transfer learning also exhibits certain drawbacks. For instance, it may yield negative results when there is a substantial disparity between the source and target domains. Moreover, the effectiveness of transfer learning depends on the extent to which knowledge can be transferred between the two domains. If the transfer of knowledge proves to be inadequate or inefficient, the performance improvements achieved through transfer learning may be insignificant.

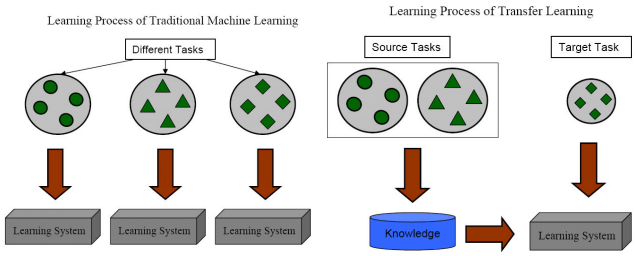


Fig. 1. Contrasting learning processes: (a) traditional machine learning versus (b) transfer learning.

2) Classification: As shown in Fig. 2, transfer learning can be divided into different categories according to different classification criteria.

Based on the distinctions between domains, transfer learning is categorized into two types: homogeneous and heterogeneous [44]. The former involves processing domains within the same feature space, where the assumption is that they primarily differ in marginal distributions. In this context, the domain adaptation is achieved by mitigating the sample selection bias [45] or covariate shift [46]. On the other hand, heterogeneous transfer learning deals with knowledge transfer when the feature spaces of each domain are different. Apart from distribution adaptation, it also necessitates feature space adaptation [47], thereby making it more intricate compared to homogeneous transfer learning.

According to different transfer approaches, transfer learning can be classified as instance-based, feature-based, parameter-based, and relational-based [43]. The instance-based transfer learning primarily utilizes the sample weighting strategy to assign weights to different samples. Feature-based transfer learning involves transforming features, particularly when the features of the source and target domains are not in the same space or exhibit heterogeneity in the original space. This transformation aims to obtain new feature representations. Parameter-based transfer learning focuses on reusing parameters in the model, particularly applicable to neural networks that can be directly transferred. Relational-based transfer learning refers to the transfer of logical relationships or rules learned in the source domain to the target domain by mining and utilizing relationships for analogy. For instance, the scenario of teachers attending classes and students attending classes can be analogized to a company meeting.

According to the specific situation of domain data, transfer learning can be divided into three types: inductive transfer learning, transductive transfer learning, and unsupervised transfer learning [48], [49]. Transductive transfer learning occurs when the label information can only be obtained from the source domain. If the label information of the target-domain instances is available, it falls under the category of inductive transfer learning. Unsupervised transfer learning refers to situations where the label information is unknown in both the source and target domains.

3) Domain Adaptation: Domain adaptation is a frequently utilized term in the context of transfer learning. It denotes the procedure of adapting to one- or multiple-source domains in

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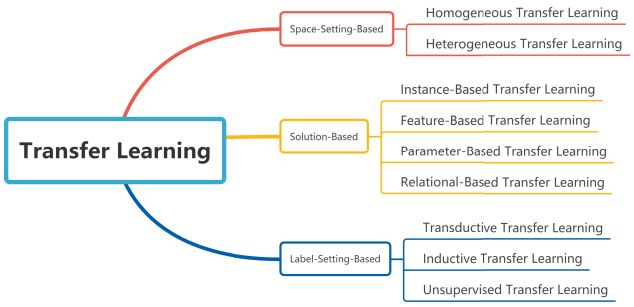


Fig. 2. Categorizations of transfer learning.

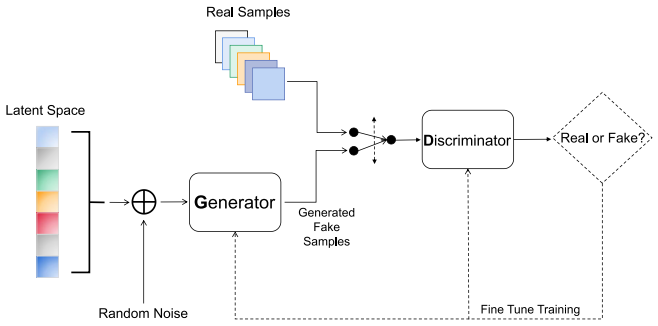


Fig. 3. Architectural diagram of GAN.

order to transfer knowledge and enhance the capabilities of the target learner [44]. Transfer learning often leans on domain adaptation processes in an attempt to minimize the domain discrepancy.

Specifically, the fundamental concept of domain adaptation is to map the data from the source and target domains, which possess the same sample space but exhibit disparate data distributions, into a feature space. The aim is to identify a metric criterion within this feature space that narrows the disparity between the feature distributions of the source- and target-domain data. In this way, the target function discriminator trained on the features from the source-domain data in the feature space can be effectively transferred to the target-domain data. In consequence, the accuracy of the target domain is significantly enhanced [43].

Existing domain adaptation approaches can be broadly classified into methods with shallow and deep architectures. Shallow domain adaptation methods [50], [51] mainly employ instance- and feature-based techniques to align the distributions between domains. One common approach is minimizing the distance between domains using various metrics, such as MMD, the Wasserstein metric, and correlation alignment (CORAL) [52]. In contrast, deep domain adaptation approaches [53], [54] leverage neural networks. These approaches typically utilize convolutional, AE, or adversarial networks to reduce the domain gap. Some methods in this category may also utilize a distance metric in one or multiple layers of two networks (one for source data and one for target data) to measure the discrepancy between their corresponding feature representations on the corresponding layers [55]. Overall, shallow domain adaptation methods focus on aligning domain distributions using specific distance measures, while deep domain adaptation methods employ neural networks to bridge the domain gap. Both approaches have demonstrated effectiveness in addressing domain shift challenges, but the deep architecture offers the advantage of leveraging more complex representations learned by deep neural networks.

B. Generative Adversarial Network

GAN is a generative model proposed by Goodfellow et al. [58] It trains the network through adversarial mode to generate new data samples that can effectively estimate the underlying distribution of the samples. It offers a means to acquire deep learning representations independently of labeled data [56].

1) Basic Idea: The idea of GAN is inspired by the zero-sum game and Nash equilibrium, where the total gains of both parties involved in the game remain constant. In the process of competition, one party gains and the other party must lose something [57]. Therefore, GAN is trained in an adversarial manner to ensure that the samples generated by the generative network closely adhere to the distribution of real data.

GAN utilizes two networks for adversarial training: the generator network and the discriminator network, as illustrated in Fig. 3. The generator network aims to simulate the distribution of real samples by incorporating noise as a latent variable. Its objective is to generate samples that are indistinguishable from real data, making it challenging for the discriminator network to detect their source. On the other hand, the goal of the discriminator network is to accurately classify whether a given sample originates from real data or was generated by the generator network. These two networks, with opposing objectives, engage in a competitive process where they continually train and update alternately until they reach a state of equilibrium [58]. In the final convergence, if the discriminator network is unable to determine the source of samples, it is equivalent to that the generator network has successfully generated samples that closely match the distribution of real data.

GANs possess remarkable capabilities, chiefly their potent generative prowess and proficiency in unsupervised learning. These advantages pave the way for data augmentation, whereby GANs can enrich training sets by generating samples that bear resemblance to, but are subtly distinct from, the original data. This process contributes significantly to enhancing the model's generalizability and robustness. However, GANs are not without their limitations. Notably, their training procedures have a tendency to be unstable, potentially leading to issues such as model collapse. Consequently, they may struggle to fully capture the intricate diversity and comprehensive detail of data distribution.

2) Algorithm: The training process of the GAN algorithm is depicted in Algorithm 1. In the algorithm, $D(x)$ represents a discriminator that takes input data and outputs the probability of the input being true. $G(z)$ is a generator that takes input data and produces data with a specific distribution, aiming

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to match the input of $D(x)$. $x^{(i)}$ is the real data, marked as one, and $G(z^{(i)})$ is the fake data generated by the generator, marked as zero.

Algorithm 1 Training Process of GAN

Input: T: the iteration number of adversarial training, K: the training iteration number of discriminator network, M: the number of minibatch samples. **Output:** Generative network $G(z)$

for t ← 1 to T do **do**
  //Training discriminator network $D(x)$
  for k ← 1 to K do **do**
    //Collecting minibatch training samples
    m samples $x^{(i)}(1 \le \mathrm{i} \le \mathrm{m})$ are collected from the data generation distribution $p_{data}(x)$
    m noise samples $z^{(i)}(1 \le \mathrm{i} \le \mathrm{m})$ are collected from the probability distribution $p_g(z)$ of the noise
    The discriminator is updated using a random gradient ascent. The gradient is:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(x^{(i)}\right) + \log\left(1 - D\left(G\left(z^{(i)}\right)\right)\right)\right]. \quad (1)$$

  end for
  //Training generator network $G(z)$
  m noise samples $z^{(i)}$ $(1 \le \mathrm{i} \le \mathrm{m})$ are collected from $p_g(z)$

  Update generator using stochastic gradient descent with gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(z^{(i)}\right)\right)\right). \quad (2)$$

end for

For the discriminator, the loss function is given by (1). It can be observed that this formula bears a resemblance to the logarithmic loss function frequently employed in neural networks and logistic regression. Specifically, when the input follows a zero-one distribution, the loss function is given as follows:

$$-\frac{1}{N} \sum_{i=1}^{N} \left(y_i \log p_i + (1 - y_i) \log(1 - p_i)\right). \quad (3)$$

It is obvious that, if the label in the logarithmic loss function for a binary classification problem is 1, the latter part becomes 0. Conversely, if the label is 0, the first half becomes 1. As a result, the loss function presented in (1) is equivalent to the logarithmic loss function. Due to the favorable mathematical properties of the logarithmic loss function, optimization algorithms, such as the gradient descent algorithm, can be employed to efficiently minimize the loss function. It is worth noting that (1) deviates from the logarithmic loss function due to the negative sign, transforming it into a maximization loss function. Consequently, the stochastic gradient ascent algorithm is adopted.

For the generator, the loss function is given by (2). When optimizing the generator, all the output results of $G(z^{(i)})$ are marked as 1 and fed into the discriminator $D(x)$. The actual error is generated by the discriminator, but this error is propagated back to the generator to adjust its parameters. Therefore, the loss function still employs the logarithmic loss function used in binary classification problems. The rationale behind utilizing the stochastic gradient descent algorithm here is that maximizing $D(G(z^{(i)}))$ is equivalent to minimizing $\log(1 - D(G(z^{(i)})))$.

In summary, the training process can be outlined as follows.

1) *For the Discriminator:*
   a) Input noise into the generator, which generates synthetic data conforming to a specific distribution (e.g., the distribution of a set of images).
   b) Select a subset of data from the generated distribution, and label it as 0; select another subset of data from the real sample set, and label it as 1.
   c) Train the discriminator using the labeled data to enhance its ability to discern between real and synthetic data.
2) *For the Generator:*
   a) Establish a logical connection between the generator and the discriminator.
   b) Label all the synthetic data generated by the generator as 1, and feed it to the discriminator.
   c) Utilize the error backpropagation algorithm to propagate the error from the discriminator back to the generator, thereby updating the generator's parameters.

The discriminator and the generator are trained iteratively in an adversarial manner, continually improving their performance. The training process persists until the discriminator reaches a point where it can no longer differentiate between real and synthetic data, resulting in an output probability of approximately fifty percent. This equilibrium signifies the convergence of the discriminator and the generator, indicating the completion of the training process.

C. Few-Shot Learning

FSL, also known as low-shot learning (LSL) in minority data, refers to a machine learning problem in which the training dataset contains only a limited amount of information. In typical machine learning applications, it is common practice to provide the model with a large amount of data as more data often lead to improved model predictions. However, in FSL, the objective is to recognize new categories with very few labeled samples [59]. The FSL aims to develop accurate machine learning models using a reduced number of training datasets. This approach holds progressive scientific significance as it contributes to reducing the consumption cost of data analysis and machine learning processes.

1) Distinct From Traditional Supervised Learning: The objective of traditional supervised learning is to enable the machine model to recognize the data in the training set and generalize its knowledge to the test set. In contrast, FSL aims to facilitate the machine's ability to learn and adapt autonomously. In traditional supervised learning, the training set is used to train a model, which is then utilized to predict test data. The test data, although not present in the training

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set, belongs to categories that are already included in the training set. However, the test data in FSL not only lack its own presence in the training set but also belong to categories that are entirely absent from the training set. This fundamental distinction between FSL and traditional supervised learning highlights the unique challenges and goals of FSL approaches.

2) Basic Idea: In FSL, the small sample training set typically comprises multiple categories, each containing a few samples. During the training stage, C classes are randomly chosen from the training set, with K samples per class. A total of $C \times K$ samples are then selected to form a metatask, which serves as the input for the model's support set and can be denoted as $S = \{(x_i, y_i)\}_{i=1}^{C\times K}$. Next, N samples are selected from the remaining data of the same C classes as the prediction objects of the model, which is called query set and can be expressed as $Q = \{(x_j, y_j)\}_{j=1}^{C\times N}$. The model is required to learn how to distinguish the C classes according to the $C \times K$ data, which is called the C-way K-shot problem [60]. Experimental results demonstrate that the prediction accuracy of FSL decreases as the number of classes (C-way) increases but increases with the increase in samples per class (K-shot). This is because the greater number of categories makes the classification more susceptible to errors. However, increasing the number of samples provides more reference information, which improves the accuracy of predictions.

It is important to note that the training set is typically large, containing numerous sample data for each category, which allows for training a deep neural network. In contrast, the support set is small and comprises only a few samples per class, which is insufficient for training a large neural network. Consequently, the support set can only provide limited additional information during testing.

During the training process, different metatasks are sampled in each training iteration, resulting in a diverse set of category combinations. This enables the model to learn the shared characteristics across different metatasks [61]. As a result, when faced with new and unknown metatasks, the model trained using this learning mechanism exhibits improved classification performance and generalization capability.

The basic idea of the FSL method is to learn a similarity function $\mathrm{sim}(x_1, x_2)$ to judge the similarity between x_1 and x_2. The similarity between two samples is directly related to the likelihood of them belonging to the same category. Thus, it is possible to learn a similarity function from a large dataset and utilize it for making predictions.

FSL exhibits notable traits of low data requirements and high efficiency, enabling task completion with a sparse set of labeled samples and rapid training cycles. Furthermore, as FSL distills shared features from a limited sample pool, its model showcases robust generalization capabilities when faced with novel samples. However, it is susceptible to overfitting and heavily reliant on domain expertise. When dealing with intricate tasks and extensive datasets, FSL might fall short of delivering optimal performance.

3) Deep Few-Shot Learning: DFSL is a method proposed by Liu et al. [35], which employs deep learning techniques to address the few-shot problem in HSI classification. This approach utilizes a deep residual 3-D CNN to extract spectral–spatial features and mitigate label uncertainty. The network is trained using episodes to learn the metric space, where samples from the same category are grouped closely, while samples from different categories are separated. During testing, the nearest neighbor classifier is employed to classify test samples based on their proximity in the learned metric space.

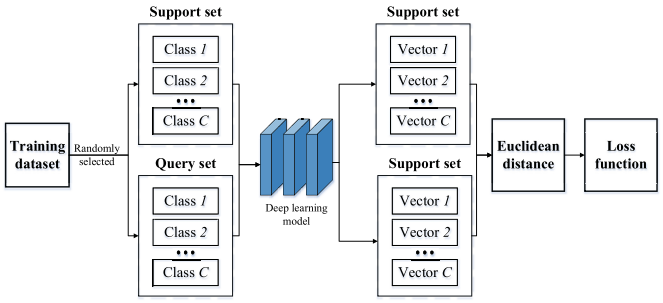


Fig. 4. Structure of the DFSL method.

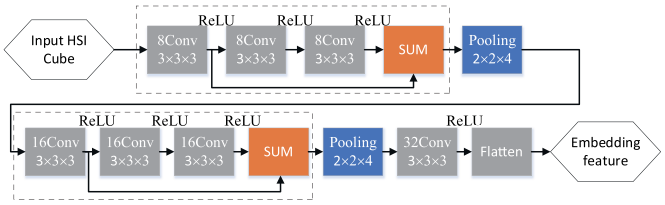


Fig. 5. Architecture of deep residual 3-D CNN.

The key idea of the DFSL algorithm is to design a network that learns a metric space from the training dataset, and this metric space can be generalized to the classes in the test dataset, including those unseen during training. The framework of the algorithm is illustrated in Fig. 4.

Subclasses of classes from the training set are randomly selected to form episodes, which are used to compute gradients and update the network parameters. The support set consists of selected samples from each class, while the remaining samples are reserved as the query set. These two sets of samples are passed through the network to extract embedded features. By calculating the softmax based on the distances of the samples in the support set in the embedding space, the class distribution of the query samples can be obtained.

The deep learning model used for extracting embedded features and parameterizing metric space is based on the deep residual 3-D CNN. This model has the capability to directly extract spectral–spatial features from the data cube without the need for any preprocessing. The architecture diagram is shown in Fig. 5.

The network is composed of two residual blocks, two pooling layers, and a convolutional network layer serving as the embedding function. To account for the change in data size caused by pooling, the pooling layer is not utilized within the residual blocks. Instead, each residual block is linked to a 3-D maximum pooling layer, serving the purpose of streamlining computation and aggregating features. Subsequently, the feature map is expanded into a 1-D vector to extract the embedded features. The network has the capability to learn a metric space

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in which samples belonging to the same category are closely clustered together.

The classification process for the test dataset consists of three key stages. Initially, the pretrained deep residual 3-D CNN is employed to extract embedded features. Subsequently, the Euclidean distance between the labeled sample and the sample under classification is computed. Finally, the NN classifier determines the ultimate label based on these calculations.

The deep residual 3-D CNN designed in this method serves as an embedding function following its training. During the classification process of the testing datasets, all instances undergo feature extraction by passing through the pretrained deep residual 3-D CNN. Subsequently, a subset of samples that have been labeled is chosen in a random manner as supervised samples. The network after training facilitates the proximity of similar samples to one another within the embedding space. As a result, the test samples can be classified using a simple NN classifier. It is essential to note that the training and testing datasets are mutually independent. Finally, the classification result maps derived from the labels of the test samples are compared against the ground-truth maps to evaluate the various classification methods.

D. Cross-Scene Classification

For an extended period, supervised learning has remained a primary research focus in HSI classification. When an ample amount of training samples is available and both the training and test data exist within the same feature space, supervised classification can yield highly satisfactory results. However, in practical remote sensing image processing, the HSI scenes that require classification (referred to as target domains) often possess only a limited number of labeled training samples due to the laborious and time-consuming nature of manual labeling. As a consequence, this limitation negatively impacts the classification accuracy. Conversely, similar scenarios (known as source domains) may contain a substantial number of labeled samples. Consequently, training the target-domain model can be assisted by leveraging the pertinent class-specific information available in the source scene, and this approach to HSI classification is commonly referred to as cross-scene classification [62].

The effectiveness of knowledge transfer in the HSI classification task is severely hindered by the distribution differences observed among HSIs captured by different sensors [63]. Consequently, the primary distinction between cross-scene HSI classification and traditional HSI classification lies in the transfer of existing knowledge. As a result, current research on cross-scene HSI classification can be broadly categorized into two major directions: transfer learning and multitask learning. Transfer learning, which has been extensively discussed in the first part of Section II, focuses on leveraging prior knowledge and transferring it to improve classification performance. Therefore, it will not be reiterated here. Multitask learning [64] is an approach that tackles multiple learning tasks simultaneously by capitalizing on the commonalities and differences across tasks. Compared to training individual models in isolation, multitask learning offers enhanced learning efficiency and prediction accuracy for specific tasks.

Cross-scene classification leverages preexisting labeled data and models from source scenarios, resulting in time and resource savings, while facilitating swift model deployment and knowledge transference. This approach widens the model's scope of application across diverse scenarios. Nevertheless, its effectiveness is not without constraints, including disparities between domains, label inconsistencies, potential negative transfer effects, and more. Furthermore, due to the potential impact of source-domain traits on model performance within the target domain, cross-scene classification may introduce a level of interpretational ambiguity into the model's predictive outcomes.

III. Proposed Method

A. Overall Framework

The flowchart of the proposed GLS is shown in Fig. 6, comprising three main components: the AE, the embedded feature extractor, and the domain discriminator. The AE is designed to map the input data into a latent space while reducing the feature dimensions of both input data samples to ensure that they have the same dimensions. Then, the embedded feature extractor, composed of a deep residual 3-D CNN, is employed to extract spatial–spectral embedded features from the latent space. Finally, a conditional adversarial domain adaptation strategy is leveraged to address domain shift, with the incorporation of MMD loss to minimize distribution differences between the two domains and facilitate better alignment of the domain distributions.

In the training stage, FSL is performed alternately using few-shot datasets from both the source and target domains. The process can be divided into four key components.

1) First, the encoder module of the AE is utilized to project the data into a latent space, ensuring that the input dimensions between the source and target domains in this space are standardized.
2) Second, the embedded feature extractor is implemented, and the loss function of MMD is introduced as a measure of the distribution difference between the source and target domains. This helps reduce the disparity between the domains and trains the spatial–spectral embedding space.
3) Next, the spatial–spectral embedding features of the data are obtained by calculating the distance between the labeled samples and the unlabeled samples in the latent space. Subsequently, FSL is conducted to leverage the obtained embedded features and enable effective classification.
4) Finally, the domain discriminator is implemented based on an adversarial network to mitigate the domain shift between the two domains and achieve the domain distribution alignment.

Due to the distinct category spaces between the source and target domains, applying one classifier to both domains may result in significant errors during classification. Consequently, this method employs two classifiers with identical structures but different parameters to perform separate training. By doing this, the source- and target-domain data can be classified

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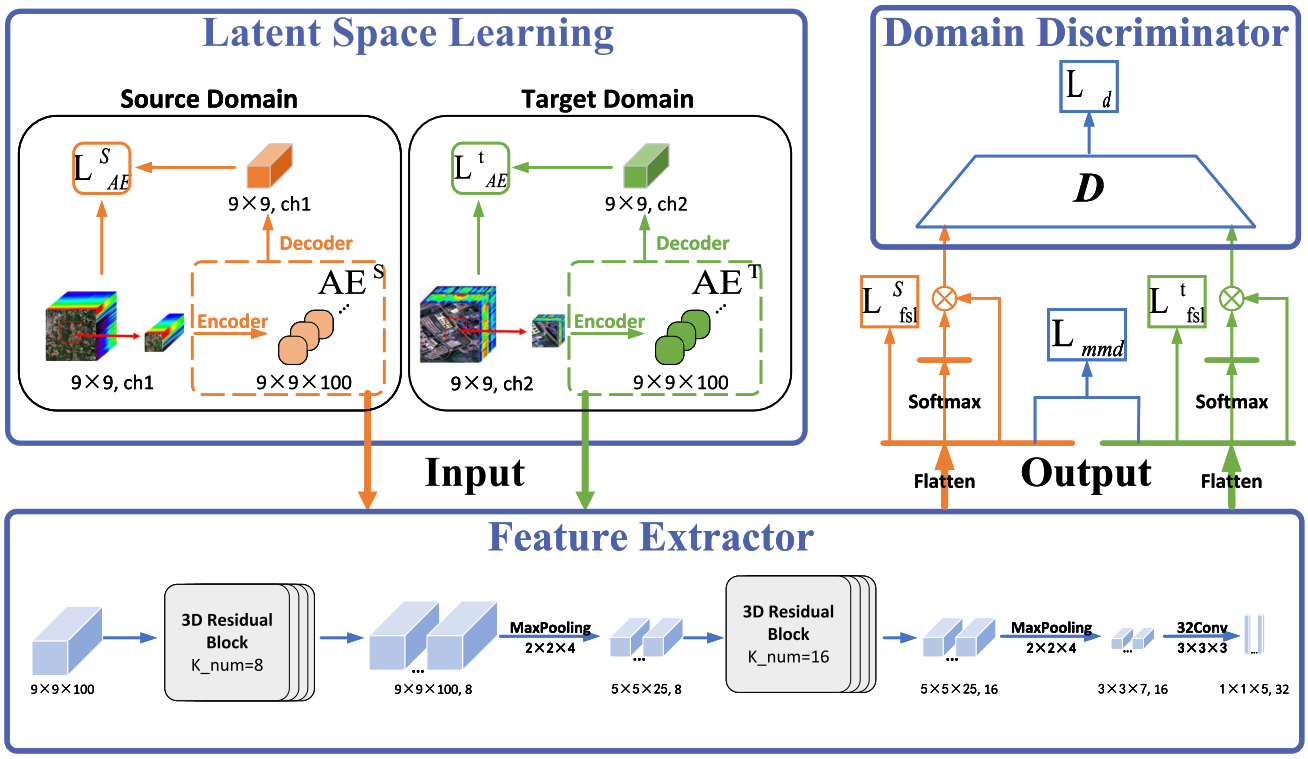


Fig. 6. Overall framework diagram. The module contains three parts: latent space learning, feature extractor, and domain discriminator. The first part utilizes AE networks to map the input data into a latent space, where dimensionally consistent data are obtained and fed into the second part, the feature extractor, to extract spectral–spatial embedded features. Finally, the third part employs a conditional adversarial domain adaptation strategy to reduce domain discrepancies while introducing MMD loss for further domain distribution alignment.

independently by training different classifiers, ensuring more accurate and reliable classification results.

During the testing stage, the unlabeled samples in the target domain undergo initial processing by the trained AE network to project them into a latent space. Subsequently, the trained deep residual 3-D CNN is utilized to extract the spatial–spectral embedding features from these samples. Finally, the neural network classifier performs the classification of the unlabeled samples, resulting in the generation of a classification map that assigns labels to the unlabeled samples.

In summary, the key innovations of this article are given as follows. First, to preserve the discriminative features, we propose a feature extraction module that utilizes AEs to map the input data to a latent space. Subsequently, a deep residual 3-D CNN is employed in this space to extract the most efficient feature representation, facilitating feature alignment and minimizing information loss. Second, in addition to the conditional adversarial domain adaptation for addressing domain shift, we incorporate the MMD loss function. This function effectively measures the distribution distance between the source domain and the target domain, reducing the disparity between the domains and enhancing interdomain adaptation. These innovations collectively contribute to the advancement of the proposed methodology.

The subsequent sections concentrate on the pivotal structures employed in the approach presented in this article.

B. AE-Based Latent Space Mapping

The AE is an unsupervised learning model that plays a crucial role in mapping the entire input space into a lower dimensional space, aiming to minimize information loss during feature dimension reduction. Given the substantial spectral information present in HSIs, computing all the spectra becomes computationally intensive, and convolutional networks require consistent input sizes. Therefore, prior to extracting spatial–spectral embedding features, this article utilizes an encoder to project the input data into a latent space. This projection ensures that both input data share the same dimensionality within this space, thereby providing feature inputs of consistent dimensions for the embedded feature extractor. The proposed latent space mapping module based on AE in this article effectively preserves more meaningful feature information during dimensionality reduction, thus enabling more efficient input for subsequent feature extraction and sample classification.

Specifically, the AE algorithm consists of two main components: the encoder and the decoder. The encoder is employed to reduce the input dimensionality between the source domain (e.g., 128 spectral bands of the Chikusei dataset) and the target domain (e.g., 204 spectral bands of the Salinas dataset) before feature extraction. Simultaneously, the decoder reconstructs the data with reduced dimensions and calculates the difference between the reconstructed image and the original image as the loss of the AE, which is then backpropagated to update the network parameters. The network structure of the AE is illustrated in Fig. 7.

The encoder module shrinks the input data to a lower dimensional representation, while the decoder module endeavors to reconstruct the original input using this compressed representation. The discrepancy between the original data

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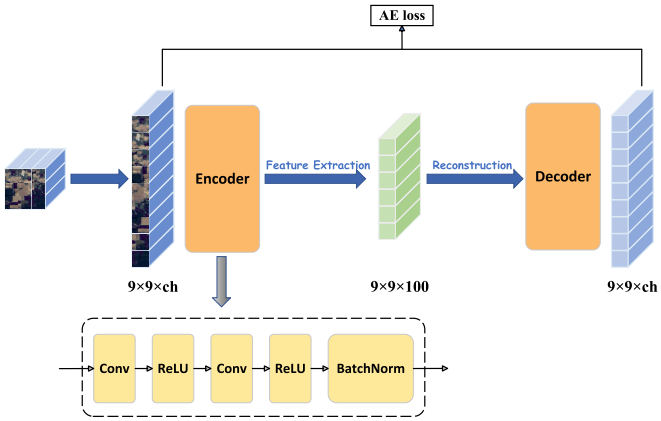


Fig. 7. AE network architecture. The network consists of an encoder and a decoder. The encoder projects the data into a latent space while reducing the dimensionality to obtain dimensionally consistent features in the latent space. The decoder is utilized to reconstruct the encoded data, minimizing the reconstruction loss to reduce information loss during the feature dimension reduction process.

and the reconstructed data is known as the reconstruction error. During the process of dimensionality reduction, some loss of information is inevitable, resulting in the reduced-dimensional features not being the most representative of the original data. To address this, the network is trained to minimize the reconstruction error on the dataset by jointly optimizing the encoder and the decoder. Through this training, the network learns to discover the most effective approach to compress the input data into a lower dimension while leveraging the inherent structure of the data. This module operates within this framework and serves this purpose.

C. Cross-Scene Classification Based on Domain Adaptation

1) Domain Confusion Method Based on MMD: In our approach, we utilize a sophisticated deep residual 3-D CNN as the embedded feature extractor to effectively train the spatial–spectral embedding space. This enables us to obtain spatial–spectral embedding features for both the data in the source domain and the target domain. These features are derived by measuring the distances between different categories of unlabeled samples and labeled samples within the embedding space. Subsequently, these features are employed for FSL.

Taking into account the disparity in domain distribution between the source- and target-domain data, this article introduces a standard distribution distance metric called MMD [65], which quantifies the difference between the source and target distributions. It applies a mapping function $\phi(x)$ to map data x_s from the source domain X_S and data x_t from the target domain X_T to other spaces. The differences between the mean values after the mapping are then described. As the feature extraction network in this experiment has already mapped the data from both the source and target domains into the same metric space, the necessity of mapping the data to other spaces is reduced. The calculation formula for the MMD loss function is given as follows:

$$\mathrm{MMD}(X_S, X_T) = \left\| \frac{1}{|X_S|} \sum_{x_s \in X_S} \phi(x_s) - \frac{1}{|X_T|} \sum_{x_t \in X_T} \phi(x_t) \right\|. \tag{4}$$

When utilizing this method to measure the distribution difference between the source and target domains for achieving domain adaptation, directly applying it to the data from both domains can result in larger classification errors due to the distinct categories of samples in each domain. It would require the characteristics of samples from these two different categories to be consistent, which is not feasible. To address this, this article employs the MMD method between the two domains. Specifically, the mean value is computed for the samples in each domain to obtain the representations of the two domains, followed by calculating the distance. This approach helps prevent an increase in classification errors and ensures consistency in dimensionality. Subsequently, the MMD loss is calculated based on the domain representations, aiming to minimize the feature differences between the source and target domains extracted by the feature extractor.

2) Domain Discriminator Based on Adversarial Network: The input x of the model is passed through the feature extractor F to obtain the feature representation f, which is then fed into two networks: the label classifier G and the domain discriminator D. The label classifier outputs the data label y, while the domain discriminator predicts the domain source of the feature representation d. The objective of G is to minimize the label classification error (FSL loss), whereas D aims to minimize the domain classification error (domain loss), meaning that it strives to distinguish between the two domains as accurately as possible. The feature extractor F provides a shared feature representation for both G and D, thereby minimizing the FSL loss (aiding G) and maximizing the domain loss (opposing D). Consequently, D acts in an antagonistic manner toward the training objectives of F and G.

The adversarial network is implemented by incorporating a gradient reversal layer (GRL) into the domain discriminator network. During backpropagation, the loss, representing the difference between predicted and actual values, is propagated backward through each layer. At each layer, the network calculates the gradient based on the error feedback, subsequently updating the parameters of that specific layer. GRL, however, multiplies the error of this layer by a negative value, effectively reversing its sign. As a result, the training objectives of the network before and after GRL become opposing, thereby achieving the desired adversarial effect [66].

The classifier G contains discriminative information and can rely on the adversarial adaptation of the feature representation f. The domain adversarial loss function L is defined on the domain discriminator D, the feature representation $f = F(x)$, and the classifier prediction $g = G(x)$

$$\min_D \max_{F,G} L = -E_{x_i^s \sim P_s(x)} \log\left[D\left(f_i^s, g_i^s\right)\right] - E_{x_j^t \sim P_t(x)} \log\left[D\left(f_j^t, g_j^t\right)\right] \tag{5}$$

where $D(,)$ represents the probability of the domain discriminator predicting that x is a sample from the source domain

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and $1-D(,)$ represents the probability of the domain discriminator predicting that x is a sample from the target domain. The goal of the domain discriminator D is to minimize the mentioned function, whereas the feature extractor F and the label classifier G aim to maximize it.

3) Connection and Difference Between the Two Methods: The domain discriminator aims to reduce the loss in domain classification, employing an adversarial approach to make it challenging for the feature extraction network to differentiate whether the feature vector originates from the source or target domain. This adversarial mechanism facilitates the reduction of differences between the source and target domains. In addition, the MMD method leverages the output features from the feature extraction network to diminish the disparities between the two domains by calculating the distribution distance, specifically the MMD loss. Therefore, both losses serve the purpose of globally reducing the domain differences.

However, despite sharing a common goal, they differ fundamentally. The domain discriminator focuses on intersample adaptation, while the MMD operates on the domain representation, which is the mean value of samples within the domain, enabling domain-to-domain adaptation as a whole.

D. Objective Function

In the FSL method, a small batch of samples is formed by randomly selecting C classes from the source class C_s, effectively creating a classification task. Within this batch, K samples are chosen from the selected class to constitute the support set $S_s = \{(x_i, y_i)\}_{i=1}^{C\times K}$, while N samples from the same class are designated as the query set $Q_s = \{(x_j, y_j)\}_{j=1}^{C\times N}$. The class distribution of the query sample x_j in the query set Q_s can be described as follows:

$$P\left(y_j = k | x_j \in Q_s\right) = \frac{\exp\left(-d\left(f_\phi\left(x_j\right), c_k\right)\right)}{\sum_{k=1}^{C} \exp\left(-d\left(f_\phi\left(x_j\right), c_k\right)\right)} \tag{6}$$

where $d(,)$ represents the Euclidean distance function and the embedded feature of the kth class in the support set is represented as c_k. C represents the number of classes in each small batch. $f(\phi)$ refers to the AE mapping layer and embedded feature extraction network, where the parameter is denoted as ϕ, and y_i represents the label of x_i.

In accordance with the negative logarithm probability of the query sample x and its actual label k, the classification loss of all query samples in a source task can be expressed as follows:

$$L_{\text{fsl}}^{s} = E_{S_s, Q_s}\left[-\sum_{(x,y)\in Q_s} \log p_\phi(y = k|x)\right]. \tag{7}$$

In the given formula, S_s and Q_s represent the support set and the query set, respectively, derived from the source-domain dataset.

Likewise, the classification loss in each target task can also be defined as follows:

$$L_{\text{fsl}}^{t} = E_{S_t, Q_t}\left[-\sum_{(x,y)\in Q_t} \log p_\phi(y = k|x)\right] \tag{8}$$

where S_t and Q_t are the support set and the query set from the target-domain dataset, respectively.

On the basis of the domain adversarial loss function L introduced above, the conditional adjustment strategy is adopted as follows:

$$T(h) = \begin{cases} T_{\otimes}(f, g), & d_f \times d_g \leq 1024 \\ T_{\odot}(f, g), & \text{otherwise} \end{cases} \tag{9}$$

where d_f and d_g are the dimensions of the f and g, respectively, $\otimes$ defined as the outer product of multiple random vectors, $\odot$ represents the product by elements, $T_{\odot}(f, g) = (1/\sqrt{d})(R_f f) \odot (R_g g)$, $R_f \in R^{d\times d_f}$, $R_g \in R^{d\times d_g}$, and $d \ll d_f \times d_g$. It can be concretely expressed as

$$\begin{aligned} L_d = \min_D \max_T L = &-E_{x_i^s \sim P_s(x)} \log\left[D\left(T\left(h_i^s\right)\right)\right] \\ &-E_{x_j^t \sim P_t(x)} \log\left[1 - D\left(T\left(h_j^t\right)\right)\right]. \end{aligned} \tag{10}$$

This formula is the loss function for learning domain-invariant embedded features.

When measuring the reconstruction error of the AE, the loss function adopted in this article is the mean square error (mse) function, which is specifically expressed as

$$L_{\text{AE}} = \frac{1}{M}\sum_{i=1}^{M} \left|x_i - \tilde{x}_i\right|^2. \tag{11}$$

The MMD loss function is (4), which has been elaborated on above.

Therefore, the total loss function of source-domain FSL with domain adaptation is

$$L_{\text{total}}^{s} = L_{\text{fsl}}^{s} + L_{\text{AE}}^{s} + \lambda_d L_d^{s} + \lambda_{\text{mmd}} L_{\text{mmd}}^{s}. \tag{12}$$

The total loss function of target-domain FSL with domain adaptation is

$$L_{\text{total}}^{t} = L_{\text{fsl}}^{t} + L_{\text{AE}}^{t} + \lambda_d L_d^{t} + \lambda_{\text{mmd}} L_{\text{mmd}}^{t} \tag{13}$$

where L_{fsl} is the error of FSL, L_{AE} is the reconstruction error of the AE, L_d is the error of the conditional domain discriminator, L_{mmd} is the distribution difference between the source domain and the target domain calculated based on the MMD, the sum of the last two terms is the total error of domain adaptation, and λ_d and λ_{mmd} are hyperparameters. Since the error of the MMD-based domain confusion method and the error of the domain discriminator both reduce the domain difference globally, the hyperparameter λ_d is set to 0.5, and λ_{mmd} is set to 0.25 in order to prevent overfitting.

IV. Experiments

The experiments conducted in this article utilize the open-source software framework PyTorch, and the data training and testing are performed within the constructed environment—Python 3.7 + CUDA11.2 + PyTorch 1.10.0 + torchvision 0.11.0. In addition to comparing with other methods, this article also conducts ablation experiments to verify the effect of different modules on classification accuracy optimization.

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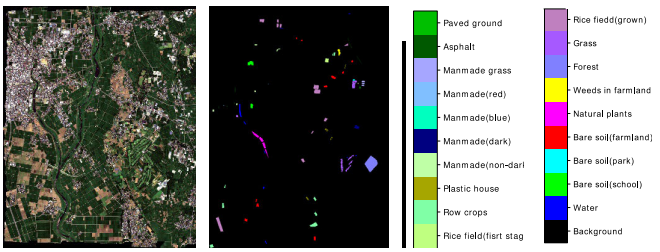


Fig. 8. Chikusei dataset. (a) False-color image. (b) Ground-truth map.

TABLE I
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE CHIKUSEI DATASET

Class	Name	Pixels
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade glass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

A. Experimental Datasets

Four public HSI datasets, namely, Chikusei, Salinas, Indian Pines (IP), and Kennedy Space Center (KSC), were employed in this experiment. Specifically, Chikusei was utilized as the source-domain dataset, while Salinas, IP, and KSC were employed as the target-domain datasets.

The Chikusei dataset contains 19 classes and has 2517 × 2335 pixels, with a spatial resolution of 2.5 m per pixel. It consists of 128 spectral bands ranging from 363 to 1018 nm. Fig. 8 displays the false-color image along with its corresponding ground-truth map, and Table I presents detailed information regarding the classes and sample quantities within the Chikusei dataset.

The Salinas dataset contains 16 classes and has 512 × 217 pixels, with a spatial resolution of 3.7 m per pixel. It consists of 204 spectral bands, ranging from 400 to 2500 nm. Fig. 9 illustrates the false-color image along with its corresponding ground-truth map, while Table II provides specific information regarding the classes and sample quantities within the Salinas dataset.

The IP dataset comprises 16 classes, with each image having dimensions of 145 × 145 pixels and a spatial resolution of 20 m per pixel. It consists of 200 spectral bands ranging from 400 to 2500 nm. Fig. 10 displays the false-color image along with its corresponding ground-truth map, while Table III presents specific information regarding the classes and sample quantities within the IP dataset.

The KSC dataset contains 13 classes and has 512 × 614 pixels, with a spatial resolution of 18 m per pixel. It consists of 176 spectral bands, ranging from 400 to 2500 nm. Fig. 11 shows the false-color image and the corresponding ground-truth map.

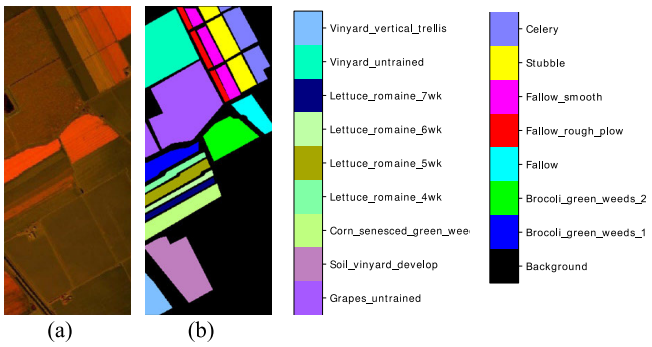


Fig. 9. Salinas dataset. (a) False-color image. (b) Ground-truth map.

TABLE II
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE SALINAS DATASET

Class	Name	Pixels
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
	Total	54129

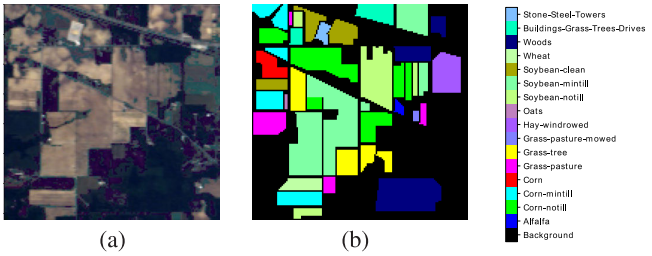


Fig. 10. IP dataset. (a) False-color image. (b) Ground-truth map.

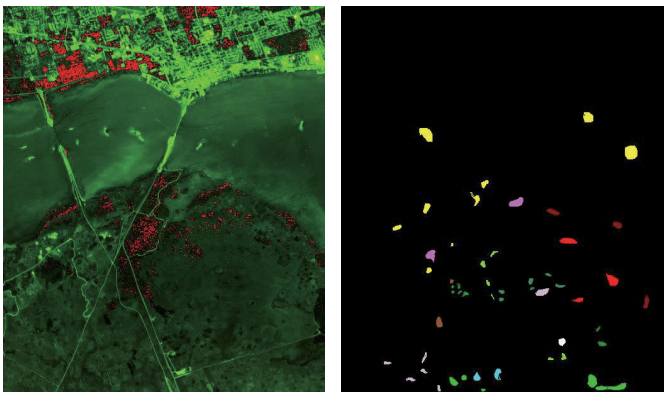
B. Experimental Results and Comparisons

To verify the effectiveness of the proposed GLS method, we conducted experiments on the Salinas, IP, and KSC datasets individually. We compared the experimental results obtained with other classification methods in both the source and

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TABLE III
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE IP DATASET

Class	Name	Pixels
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-tree	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	10249



(a) (b)

Fig. 11. KSC dataset. (a) False-color image. (b) Ground-truth map.

target domains. This comparison allowed us to evaluate the performance of our method in relation to existing approaches.

To ensure a fair comparison of the experimental results, this article adopts the same experimental parameter settings as the other methods being compared. Specifically, five labeled samples are selected from each class as supervised samples in the target domain. The number of training iterations is set to 10 000, the learning rate is set to 0.001, and all experiments are conducted ten times. The classification performance of all methods is evaluated using overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The average of the results from the ten experiments is used as the final evaluation of the method.

Tables IV–VI present the OA, AA, kappa coefficients, and classification accuracy for each category obtained from the classification experiments on the Salinas, IP, and KSC datasets, respectively. Based on these results, we can analyze the following key points.

1) The classification accuracy of the deep learning method (3-D CNN) consistently outperforms that of the standard SVM method. This is primarily due to the fact that the SVM method solely leverages spectral information from the HSI and disregards spatial information. On the other hand, the deep learning method extracts deep spectral–spatial combination features through a hierarchical network architecture, which leads to improved classification performance. The incorporation of spatial information in the deep learning approach contributes to its superior accuracy.
2) The superior classification accuracy of FSL methods (DFSL + SVM and DCFSL) compared to deep learning methods can be attributed to their ability to leverage transferable knowledge from the labeled source class data to effectively train the classifier for the target-domain data, especially in scenarios where labeled samples are limited. FSL methods excel at utilizing this transferable knowledge, enabling them to achieve better classification accuracy in such scenarios.
3) The proposed GLS method in this article exhibits higher classification accuracy compared to other FSL methods. The network architecture proposed in this study allows for the extraction of the most effective feature representation, effectively addressing domain shift through the conditional adversarial domain adaptation strategy. This adaptation strategy significantly enhances the classification accuracy of cross-domain classification tasks. Moreover, the incorporation of the MMD loss facilitates domain-to-domain adaptation by minimizing the distribution difference between source- and target-domain features extracted by the feature extractor. The synergistic effect of these modules enables the GLS network to extract more robust and domain-invariant spatial–spectral features, effectively aligning domain distributions and addressing the challenge of limited labeled samples.

By analyzing the detailed classification accuracy of different methods on the Salinas dataset in Table IV, we can find that: the DCFSL method improves OA by 2.39%, AA by 3.96%, and kappa coefficient by 2.66% compared with the DFSL + SVM method (without domain adaptation). The GLS method improves OA by 2.7%, AA by 1.65%, and kappa coefficient by 2.97% compared with the DCFSL method, respectively, which implies that the proposed model has better classification performance than other models and fully demonstrates the superiority of the proposed method.

Figs. 12–14 show the classification maps of Salinas, IP, and KSC under different methods, respectively. From a visual inspection of these maps, it is evident that the proposed method in this article achieves higher classification accuracy compared to other methods. Furthermore, the classified maps closely resemble the ground-truth maps, providing further evidence of the superiority of the proposed method.

C. Ablation Experiment

Ablation experiments serve as the control variate method in traditional experimental approaches. In this study, we conducted ablation experiments to verify the impact of the MMD loss, which was introduced in this article to minimize the distribution difference between the two domains. By comparing the classification accuracy of the GLS method before and

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TABLE IV
CLASSIFICATION RESULTS (%) ON THE SALINAS DATASET

Class	SVM	3-D-CNN	DFSL+SVM	DCFSL	GLS
1	97.57	95.29	73.92	99.40	**99.89**
2	87.43	97.20	96.85	**99.76**	99.65
3	82.95	91.45	**96.28**	91.96	96.12
4	99.11	97.31	99.11	99.55	**99.94**
5	**94.29**	91.24	80.72	92.70	93.74
6	98.36	98.80	91.63	99.52	**99.80**
7	94.39	**99.69**	97.73	98.88	99.14
8	59.99	66.40	82.33	74.57	**86.15**
9	96.09	96.25	94.44	99.59	**99.94**
10	71.45	70.72	80.96	86.42	**89.99**
11	91.25	93.15	93.38	96.61	**99.64**
12	97.22	99.65	97.94	**99.93**	98.92
13	97.30	92.63	95.79	**99.30**	98.54
14	91.84	93.56	98.87	98.85	**99.23**
15	60.52	68.02	71.13	**75.38**	71.94
16	81.45	81.41	90.57	92.22	**98.42**
OA	80.71±2.75	84.20±2.62	86.95±1.30	89.34±2.19	**92.04±0.61**
AA	87.58±1.84	89.56±1.79	90.08±1.44	94.04±1.14	**95.69±0.41**
Kappa	78.61±3.0	82.46±2.90	85.51±1.42	88.17±2.40	**91.14±0.68**

TABLE V
CLASSIFICATION RESULTS (%) ON THE IP DATASET

Class	SVM	3-D-CNN	DFSL+SVM	DCFSL	GLS
1	72.20	95.12	**96.75**	95.37	96.59
2	34.27	37.70	36.38	**43.26**	31.91
3	39.18	19.77	38.34	**57.95**	50.98
4	50.34	32.51	77.16	80.60	**89.87**
5	69.75	**88.45**	73.92	72.91	79.52
6	66.36	73.65	86.25	87.96	**95.30**
7	89.13	81.82	97.10	99.57	**100**
8	68.73	53.35	81.82	86.26	**98.31**
9	86.67	**100**	75.56	99.33	**100**
10	37.49	41.35	52.22	62.44	**66.11**
11	33.96	66.71	59.96	62.75	**69.44**
12	31.43	37.40	36.56	**48.72**	48.35
13	86.50	85.71	98.00	**99.35**	99.00
14	62.93	62.57	84.63	85.40	**89.22**
15	28.08	56.42	**74.10**	66.69	72.41
16	90.91	90.36	**100**	97.61	95.57
OA	45.85±2.44	54.76±0.03	61.69±1.85	66.81±2.37	**68.87±1.51**
AA	59.24±1.36	63.93±0.02	73.05±0.84	77.89±0.86	**80.16±1.74**
Kappa	39.68±2.48	48.72±0.03	56.78±1.90	62.64±0.86	**64.74±1.66**

TABLE VI
CLASSIFICATION RESULTS (%) ON THE KSC DATASET

Class	SVM	3-D-CNN	DFSL+SVM	DCFSL	GLS
1	48.48	73.41	**97.38**	97.29	96.53
2	48.74	66.81	89.21	84.33	**91.55**
3	43.07	68.69	83.87	85.70	**92.39**
4	17.25	29.35	**62.99**	49.11	49.03
5	41.60	53.33	48.54	71.92	**75.90**
6	18.66	43.57	70.75	92.14	**92.37**
7	46.50	81.90	65.05	96.00	**97.80**
8	30.10	47.98	77.36	86.67	**88.73**
9	79.57	79.05	**93.80**	91.28	88.12
10	36.74	49.45	**99.71**	95.44	95.56
11	85.27	89.06	**100**	95.85	96.11
12	25.92	68.48	**95.85**	93.45	91.61
13	97.89	89.59	99.37	**100**	**100**
OA	56.35±3.23	69.30±1.15	88.08±1.88	91.17±1.18	**91.59±1.32**
AA	48.11±3.07	64.67±1.79	83.38±1.96	87.63±1.58	**88.90±1.31**
Kappa	51.35±3.50	65.92±1.25	86.76±2.08	90.17±1.32	**90.64±1.47**

after incorporating MMD, we demonstrate the effectiveness of MMD in aligning domain distributions. The classification accuracies of the ablation experiments on the Salinas, IP, and KSC datasets are presented in Tables VII–IX, respectively.

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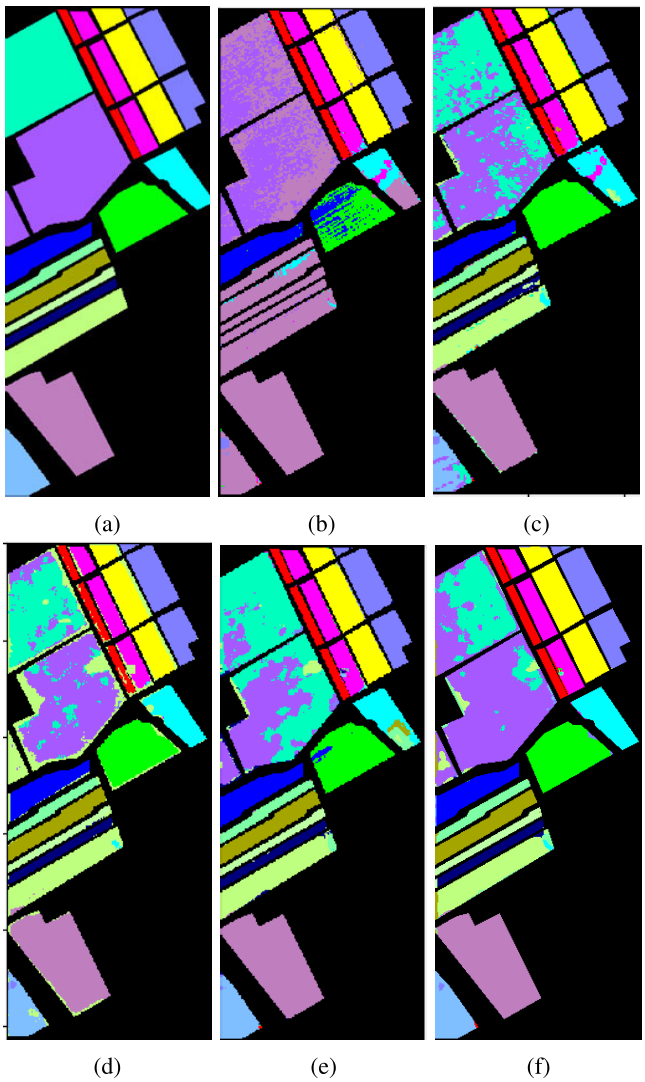


Fig. 12. Salinas dataset. (a) Ground-truth map. (b)–(f) Classification maps for different methods. (b) SVM. (c) 3-D CNN. (d) DFSL + SVM. (e) DCFSL. (f) GLS.

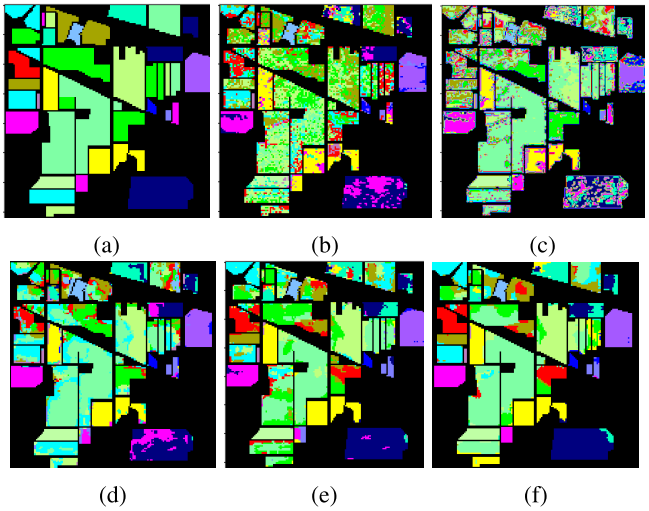


Fig. 13. IP dataset. (a) Ground-truth map. (b)–(f) Classification maps for different methods. (b) SVM. (c) 3-D CNN. (d) DFSL + SVM. (e) DCFSL. (f) GLS.

Taking the Salinas dataset as an example, the introduction of the MMD loss resulted in a significant increase in OA, AA, and Kappa coefficients by 2.18%, 0.74%, and 2.4%, respectively. This notable improvement in classification accuracy effectively validates the effectiveness and necessity of incorporating the loss function. In addition, the classification maps resulting from the ablation experiments are depicted in Figs. 15–17.

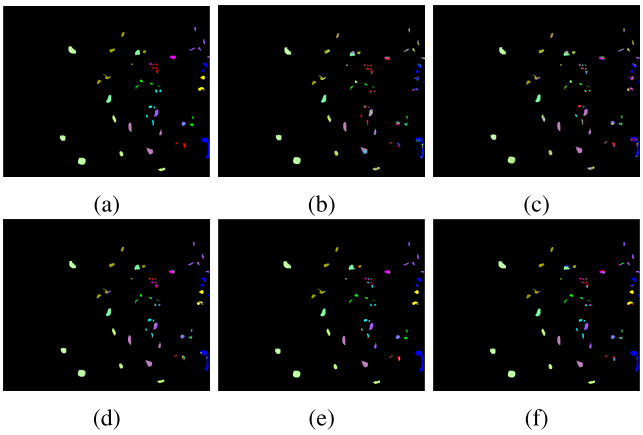


Fig. 14. KSC dataset. (a) Ground-truth map. (b)–(f) Classification maps for different methods. (b) SVM. (c) 3-D CNN. (d) DFSL + SVM. (e) DCFSL. (f) GLS.

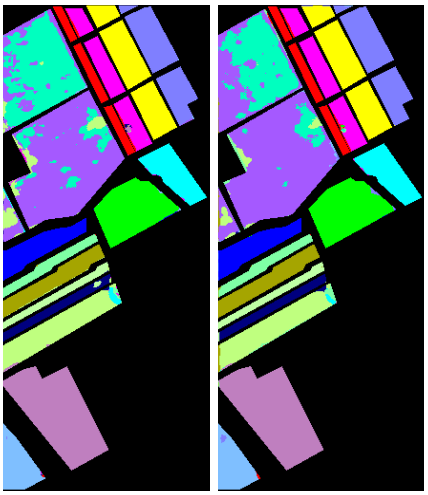


Fig. 15. Ablation experiment on the Salinas dataset. (a) GLS without MMD. (b) GLS with MMD.

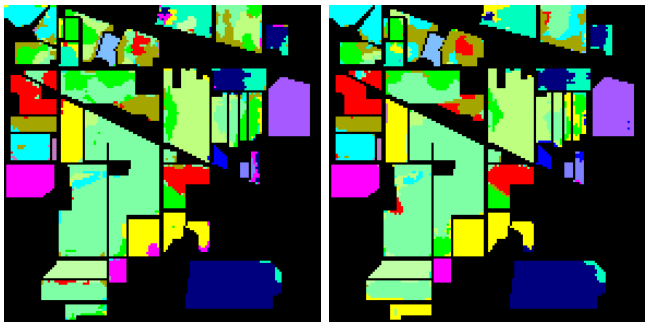


Fig. 16. Ablation experiment on the IP dataset. (a) GLS without MMD. (b) GLS with MMD.

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TABLE VII

ABLATION EXPERIMENT CLASSIFICATION RESULTS (%) ON THE SALINAS DATASET

Experiment	class1	class2	class3	class4	class5	class6	class7	class8	class9
No-MMD	98.76	99.59	95.17	99.85	92.93	99.98	99.52	76.54	99.42
With-MMD	99.89	99.65	96.12	99.94	93.74	99.80	99.14	86.15	99.94
class10	class11	class12	class13	class14	class15	class16	OA	AA	Kappa
89.98	99.15	99.87	98.93	99.20	71.37	98.88	89.86±1.58	94.95±0.56	88.74±1.75
89.99	99.64	98.92	98.54	99.23	71.94	98.42	**92.04±0.61**	**95.69±0.41**	**91.14±0.68**

TABLE VIII

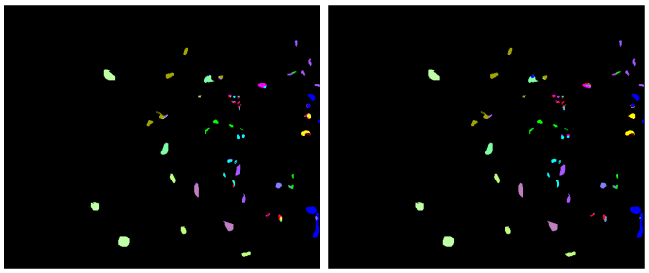
ABLATION EXPERIMENT CLASSIFICATION RESULTS (%) ON THE IP DATASET

Experiment	class1	class2	class3	class4	class5	class6	class7	class8	class9
No-MMD	96.10	30.54	48.65	91.38	78.01	95.61	100	96.57	99.33
With-MMD	96.59	31.91	50.98	89.87	79.52	95.30	100	98.31	100
class10	class11	class12	class13	class14	class15	class16	OA	AA	Kappa
63.52	71.05	41.63	99.35	90.28	66.69	96.48	68.08±0.94	79.08±1.22	63.76±1.10
66.11	69.44	48.35	99.00	89.22	72.41	95.57	**68.87±1.51**	**80.16±1.74**	**64.74±1.66**

TABLE IX

ABLATION EXPERIMENT CLASSIFICATION RESULTS (%) ON THE KSC DATASET

Experiment	class1	class2	class3	class4	class5	class6	class7	class8
No-MMD	96.89	88.66	89.84	50.00	77.48	91.61	97.10	87.70
With-MMD	96.53	91.55	92.39	49.03	75.90	92.37	97.80	88.73
	class9	class10	class11	class12	class13	OA	AA	Kappa
No-MMD	89.50	96.19	96.28	89.50	100	91.33±1.43	88.50±1.61	90.35±1.60
With-MMD	88.12	95.56	96.11	91.61	100	**91.59±1.32**	**88.90±1.31**	**90.64±1.47**



(a) (b)

Fig. 17. Ablation experiment on the KSC dataset. (a) GLS without MMD. (b) GLS with MMD.

V. CONCLUSION

In this article, we propose a cross-scene classification method of HSIs based on GLSs. Our method utilizes AE layers to map the input data to a latent space, enabling feature extraction and dimensionality reduction for both the source- and target-domain data. Furthermore, a deep residual 3-D CNN is designed in this latent space to preserve the most effective feature representation and minimize information loss during the feature extraction process. To address the domain shift challenge, we leverage conditional adversarial domain adaptation, which helps overcome the differences between domains. In addition, we incorporate the MMD loss function to reduce the distribution differences between the two domains, facilitating better alignment of domain distributions. Experimental results on three different target-domain datasets demonstrate the superiority of the proposed method. Our approach not only outperforms other deep learning methods when dealing with limited labeled samples but also surpasses existing FSL methods.

One potential avenue for future improvement in this article is to explore the use of a parameter-sharing mechanism instead of the introduced MMD loss. While the current approach reduces the error arising from the significant variability in the category space of the source and target domains by utilizing the sample mean as the domain representation and calculating the distribution distance, employing a shared parameter mechanism in the feature extraction network could further minimize this error and enhance interdomain variance reduction. In addition, this article utilizes the basic AE to map the latent space, and future work can explore advanced variations, such as the masked AE (MAE) and SAE. The MAE improves feature selection and sparsity by manipulating masks and incorporating sparsity constraints. The SAE enhances AEs' capabilities for nonlinear modeling and parameter initialization through its deep structure and layer-by-layer pretraining. These advancements may enable MAE and SAE to outperform traditional AE models in specific tasks, offering superior performance and more effective feature representation. Therefore, it would be worthwhile to investigate this method in subsequent studies.

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Yahan Yang was born in Shanxi, China, in 2001. She received the B.Sc. degree in intelligent science and technology from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China, in 2022, where she is currently pursuing the M.Sc. degree.

Her research interests include hyperspectral image classification and transfer learning.



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from the Nanjing University of Science and Technology (NUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, NUST. His research interests include hyperspectral image classification, hyperspectral detection, image processing, and machine learning.



Zebin Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2007, respectively.

He was a Visiting Scholar with the GIPSA-Lab, Grenoble Institute of Technology (Grenoble INP), Université Grenoble Alpes, Grenoble, France, from August 2018 to September 2018. He was a Visiting Scholar with the Department of Mathematics, University of California at Los Angeles, Los Angeles, CA, USA, from August 2016 to September 2016 and July 2017 to August 2017. He was a Visiting Scholar with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica Nacional, University of Extremadura, Cáceres, Spain, from June 2014 to June 2015. He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. His research interests include hyperspectral image processing, parallel computing, and big data processing.



Biqi Wang received the B.Admin. degree in information management and information systems from the School of Information Engineering, Nanjing University of Finance and Economics, Nanjing, China, in 2019. She is currently pursuing the Ph.D. degree with the Nanjing University of Science and Technology, Nanjing.

Her research interests are transfer learning and hyperspectral image processing.



Zhihui Wei (Member, IEEE) was born in Jiangsu, China, in 1963. He received the B.Sc. and M.Sc. degrees in applied mathematics and the Ph.D. degree in communication and information systems from South East University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently a Professor and a Doctoral Supervisor with the Nanjing University of Science and Technology (NUST), Nanjing. His research interests include partial differential equations, mathematical image processing, multiscale analysis, sparse representation, and compressive sensing.

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